JESS Thermodynamic Database v8.9

Tantalum

24-Jan-23

Reaction No. 40400

Ta+5_OH-1(2) + Oxalic-2 = Ta+5_OH-1(2)_Oxalic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1.5	Self Medium	lgK:11.1(3SF)	2	MSL	30107
Note (1): Ta+5_OH-1(2) improbable species							
2	18:20			lgK:11.10(4SF)	2	MSL	931
3	25	0	Inf. Dilution	lgK:11.1(3SF)	1	ELC	

Reaction No. 40401

Ta+5_OH-1(2) + 2<Oxalic-2> = Ta+5_OH-1(2)_Oxalic-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20			lgK:18.52(4SF)	2	MSL	931
2	25	0	Inf. Dilution	lgK:18.5(3SF)	1	ELC	

Reaction No. 40402

Ta+5_OH-1(2)_Oxalic-2 + OH-1 = Ta+5_OH-1(3)_Oxalic-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1.5	Self Medium	lgK:13.3(3SF)	2	MSL	30107
Note (1): Ta+5_OH-1(2) improbable species							
2	18:20			lgK:13.33(4SF)	2	MSL	931
3	25	0	Inf. Dilution	lgK:13.4(3SF)	1	ESC	

Reaction No. 43264

Ta+5_OH-1(4) + Tartaric-2 = Ta+5_OH-1(4)_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20:22	3	Unknown	lgK:0.72(0.04SD)	4	MDS	906
Note (1): Ref: G.N. Shabanova, N.A. Skovik Z. obshchei Khim 1972 42 204 but this reference does not contain these data							
2	25	0	Inf. Dilution	lgK:1.0(2SF)	1	EES	

Reaction No. 43265

Ta+5_OH-1(4) + H+1(-1)_Tartaric-2 = Ta+5_H+1(-1)_OH-1(4)_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

doi: 10.5281/zenodo.7700024

1	20:22	3	Unknown	lgK:8.83(0.08SD)	4	MDS	906
Note (1): Ref: G.N. Shabanova, N.A. Skovik, Z. obshchei Khim 1972, 42, 204 but this reference does not contain these data							
2	25	0	Inf. Dilution	lgK:9.(1SF)	1	EES	

Reaction No. 43266

Ta+5_OH-1(4) + H+1(-2)_Tartaric-2 = Ta+5_H+1(-2)_OH-1(4)_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20:22	3	Unknown	lgK:15.3(0.2SD)	4	MDS	906
Note (1): Ref: G.N. Shabanova, N.A. Skovik, Z. obshchei Khim 1972, 42, 204 but this reference does not contain these data							
2	25	0	Inf. Dilution	lgK:15.5(3SF)	1	EES	

Reaction No. 43704

Ascorbic-2 + Ta+5 = Ta+5_Ascorbic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:9.5(2SF)	0	MSP	906
2	23	0	Unknown	lgK:9.5(2SF)	0	MSP	30601
Note (2): Ta+5 unreal species							

Reaction No. 44881

Ta+5_OH-1(2) + EDTA-4 = Ta+5_OH-1(2)_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1.08	K2SO4	lgK:33.6(3SF)	0	MPL	906
Note (1): Error in reacting species. Ta+5_OH-1(5) in original publication							

Reaction No. 53173

F-1 + Ta+5 = Ta+5_F-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:6.37(3SF)	0	MIS	5766
Note (1): Ta+5 unreal species; Ta+5_F-1 improbable species							

Reaction No. 53174

2<F-1> + Ta+5 = Ta+5_F-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:11.85(4SF)	0	MIS	5766
Note (1): Ta+5 unreal species; Ta+5_F-2 improbable species							

Reaction No. 53175

3<F-1> + Ta+5 = Ta+5_F-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:16.03(4SF)	0	MIS	5766

Note (1): Ta+5 unreal species

Reaction No. 53176

4<F-1> + Ta+5 = Ta+5_F-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:19.63(4SF)	0	MIS	5766

Note (1): Ta+5 unreal species

Reaction No. 53177

5<F-1> + Ta+5 = Ta+5_F-1(5)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:23.29(4SF)	0	MIS	5766

Note (1): Ta+5 unreal species

Reaction No. 53178

7<F-1> + Ta+5 = Ta+5_F-1(7)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:30.21(4SF)	0	MIS	5766

Note (1): Ta+5 unreal species

Reaction No. 53727

Ta+5_F-1(3) + F-1 = Ta+5_F-1(4)

Note: Reacting species unknown, probably hydroxylated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:5.90(3SF)	2	MQH	5398
2	25	3	HClO4	lgK:5.86(3SF)	2	MDS	931
3	25	3	HClO4	lgK:5.86(3SF)	2	MDS	30162
4	25	3	NaClO4	lgK:5.86(3SF)	2	MDS	437

Reaction No. 54409

Ta+5_F-1(5) + F-1 = Ta+5_F-1(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0.25	HNO3	lgK:3.16(3SF)	3	MSL	437

2	25	1	(H,Na)ClO4	lgK:3.6(2SF)	4	MAX	140
3	25	1	HClO4	lgK:3.62(3SF)	4	MAX	21702
4	25	1	NaClO4	lgK:3.6(2SF)	3	MDS	437
5	25	1	NaClO4	lgK:3.75(3SF)	3	MQH	437
6	25	1	NaClO4	lgK:3.75(3SF)	3	MQH	931
7	25	1	NaClO4	lgK:3.75(3SF)	3	MQH	30109
8	25	3	HClO4	lgK:4.90(3SF)	3	MDS	437

Reaction No. 54410

Ta+5_F-1(6) + F-1 = Ta+5_F-1(7)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO4	lgK:3.3(2SF)	4	MAX	140
2	25	1	HClO4	lgK:3.34(3SF)	4	MAX	21702
3	25	1	NaClO4	lgK:3.3(2SF)	3	MAX	437
4	25	1	NaClO4	lgK:3.10(3SF)	3	MQH	437
5	25	1	NaClO4	lgK:3.10(3SF)	3	MQH	931
6	25	1	NaClO4	lgK:3.10(3SF)	3	MQH	30109
7	25	3	NaClO4	lgK:4.48(3SF)	3	MDS	437

Reaction No. 54411

Ta+5_F-1(7) + F-1 = Ta+5_F-1(8)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO4	lgK:3.0(2SF)	4	MAX	140
2	25	1	HClO4	lgK:2.97(3SF)	4	MAX	21702
3	25	1	NaClO4	lgK:0.66(2SF)	0	MQH	931
4	25	1	NaClO4	lgK:0.66(2SF)	0	MQH	30109

Reaction No. 54484

Ta+5_F-1(4) + F-1 = Ta+5_F-1(5)

Note: Reacting species unknown, probably hydroxylated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO4	lgK:4.8(2SF)	2	MAX	140
2	25	1	HClO4	lgK:4.81(3SF)	2	MAX	21702
3	25	1	NaClO4	lgK:4.8(2SF)	2	MAX	437
4	25	3	HClO4	lgK:4.91(3SF)	2	MDS	437

Reaction No. 58550

Ta+5_F-1(8) + F-1 = Ta+5_F-1(9)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO4	lgK:3.6(2SF)	3	MAX	140
2	25	1	HClO4	lgK:3.57(3SF)	4	MAX	21702

Reaction No. 66379							
Cl2(g) + 0.5<O2(g)> + Ta(s) = TaOCl2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-191.(3SF)	2	CRV	22971
2	25	0	Inf. Dilution	dH:-213.(3SF)	2	CRV	22971

Reaction No. 66380							
Ta(s) + 2<S(s)> = Ta+4_S-2(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:60.43(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:60.05(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:44.62(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:35.23(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:28.92(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-82.44(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-84.6(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-85.7(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-86.3(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-86.8(3SF)	0	MTD	6422

Reaction No. 66381							
Ta(s) + O2(g) + 0.5<Cl2(g)> = TaO2Cl(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:169.7(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:168.5(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:123.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:95.82(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:77.71(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-231.5(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-250.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-249.4(4SF)	3	MTD	6422
9	227	0	Inf. Dilution	dH:-248.8(4SF)	2	MTD	6422

10	327	0	Inf. Dilution	dH:-248.2(4SF)	0	MTD	6422

Reaction No. 66382							
Ta(s) + 0.5<O2(g)> + 1.5<Cl2(g)> = TaOCl3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:140.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:139.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:100.9(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:77.67(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:62.27(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-191.8(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-213.3(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-212.6(4SF)	2	MTD	6422
9	227	0	Inf. Dilution	dH:-211.8(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-211.0(4SF)	0	MTD	6422

Reaction No. 66383							
2.5<O2(g)> + 2<Ta(s)> = Ta2O5(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:334.8(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:332.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:243.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:190.2(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:154.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-456.7(4SF)	1	EFE	6422
7	25	0	Inf. Dilution	dG:-457.0(4SF)	4	CRV	22971
8	25	0	Inf. Dilution	dG:-460.2(4SF)	2	CRV	28287
9	25	0	Inf. Dilution	dG:-456.5(0.6SD)	4	EFE	30139
10	25	0	Inf. Dilution	dH:-489.0(4SF)	2	MTD	6422
11	25	0	Inf. Dilution	dH:-489.2(4SF)	4	CRV	22971
12	25	0	Inf. Dilution	dH:-488.8(0.5SD)	4	MCL	30139
13	127	0	Inf. Dilution	dH:-488.6(4SF)	1	MTD	6422
14	227	0	Inf. Dilution	dH:-488.1(4SF)	0	MTD	6422
15	327	0	Inf. Dilution	dH:-487.4(4SF)	0	MTD	6422

Reaction No. 66384
Ta(s) + 2.5<I2(s)> = Ta+5_I-1(5)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.90(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:51.58(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:38.64(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:28.86(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:21.25(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-70.8(3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-70.0(3SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-79.7(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-105.1(4SF)	1	MTD	6422
10	327	0	Inf. Dilution	dH:-103.7(4SF)	0	MTD	6422

Reaction No. 66385							
Ta(s) + 2.5<F2(g)> = Ta+5_F-1(5)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:313.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:311.7(4SF)	0	ENS	6422
3	95	0	Inf. Dilution	lgK:250.5(4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-428.0(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dG:-428.(3SF)	4	CRV	22971
6	25	0	Inf. Dilution	dH:-454.9(4SF)	4	MTD	6422
7	25	0	Inf. Dilution	dH:-455.0(4SF)	4	CRV	22971
8	95	0	Inf. Dilution	dH:-454.0(4SF)	4	MTD	6422

Reaction No. 66386							
Ta(s) + 2.5<Cl2(g)> = Ta+5_Cl-1(5)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:130.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:129.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:92.52(4SF)	2	ENS	6422
4	216.7	0	Inf. Dilution	lgK:72.10(4SF)	1	ENS	6422
5	25	0	Inf. Dilution	dG:-178.4(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dG:-179.(3SF)	4	CRV	22971
7	25	0	Inf. Dilution	dH:-205.3(4SF)	2	MTD	6422
8	25	0	Inf. Dilution	dH:-205.5(4SF)	4	CRV	22971
9	127	0	Inf. Dilution	dH:-204.4(4SF)	1	MTD	6422

10	216.7	0	Inf. Dilution	dH:-203.7(4SF)	0	MTD	6422

Reaction No. 66387							
Ta(s) + 2<Cl2(g)> = Ta+4_Cl-1(4)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:107.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:106.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:76.27(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:58.05(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:45.96(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-146.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-167.7(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-167.0(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-166.3(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-165.6(4SF)	0	MTD	6422

Reaction No. 66388							
Ta(s) + 1.5<Cl2(g)> = Ta+3_Cl-1(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:84.10(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:83.51(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:59.77(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:45.59(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:36.17(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-114.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-130.5(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-130.1(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-129.6(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-129.1(4SF)	0	MTD	6422

Reaction No. 66389							
Ta(s) + 1.25<Cl2(g)> = TaCl2.5(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:74.54(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:74.03(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:53.20(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:40.75(4SF)	1	ENS	6422

5	327	0	Inf. Dilution	lgK:32.47(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-101.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-114.5(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-114.1(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-113.8(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-113.4(4SF)	0	MTD	6422

Reaction No. 66390

2.5<Br2(l)> + Ta(s) = Ta+5_Br-1(5)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:95.84(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:95.20(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:67.17(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:49.71(4SF)	0	ENS	6422
5	240	0	Inf. Dilution	lgK:47.95(4SF)	2	ENS	6422
6	25	0	Inf. Dilution	dG:-130.8(4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dG:-132.(3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-598.3(4SF)kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-143.0(4SF)	3	MTD	6422
10	25	0	Inf. Dilution	dH:-143.0(4SF)	4	CRV	22971
11	127	0	Inf. Dilution	dH:-160.4(4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-159.0(4SF)	0	MTD	6422
13	240	0	Inf. Dilution	dH:-158.8(4SF)	0	MTD	6422

Reaction No. 67233

2<B(s)> + Ta(s) = TaB2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.19(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:35.97(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:26.86(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:21.40(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:17.75(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-49.37(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-50.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-50.0(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-50.0(3SF)	4	MTD	6422

10	327	0	Inf. Dilution	dH:-50.0(3SF)	0	MTD	6422

Reaction No. 67234

C(s) + Ta(s) = TaC(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.99(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:24.84(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:18.57(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:14.82(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:12.33(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-34.09(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-35.0(3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-34.44(4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-35.5(3SF)	3	CRV	22971
10	25	0	Inf. Dilution	dH:-38.5(0.6SD)	3	MCL	30139
11	127	0	Inf. Dilution	dH:-34.36(4SF)	3	MTD	6422
12	227	0	Inf. Dilution	dH:-34.27(4SF)	3	MTD	6422
13	327	0	Inf. Dilution	dH:-34.19(4SF)	0	MTD	6422

Reaction No. 67235

C(s) + 2<Ta(s)> = Ta2C(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.12(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:36.89(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:27.61(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:22.04(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:18.34(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-50.64(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-51.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-50.9(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-50.8(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-50.8(3SF)	0	MTD	6422

Reaction No. 67236

0.5<N2(g)> + Ta(s) = TaN(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.69(4SF)	4	ENS	6422

2	27	0	Inf. Dilution	lgK:39.42(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:28.45(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:21.88(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:17.53(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-54.15(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-53.(2SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-60.30(4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-60.(2SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-60.17(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-59.95(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-59.68(4SF)	0	MTD	6422

Reaction No. 67237

0.5<N2(g)> + 2<Ta(s)> = Ta2N(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.73(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:42.44(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:30.63(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:23.56(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:18.86(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-58.30(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-64.90(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-64.79(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-64.60(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-64.37(4SF)	0	MTD	6422

Reaction No. 68864

Ta+5_H2O2 + F-1 = Ta+5_H2O2_F-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(l):Water 20:80Vol	lgK:2.39(3SF)	0	MSP	6680

Reaction No. 68865

Ta+5_H2O2 + 2<F-1> = Ta+5_H2O2_F-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(l):Water 20:80Vol	lgK:2.35(3SF)	0	MSP	6680

Reaction No. 68866							
Ta+5_H2O2 + 3<F-1> = Ta+5_H2O2_F-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(l):Water 20:80Vol	lgK:1.93(3SF)	0	MSP	6680

Reaction No. 68867							
Ta+5_H2O2 + 4<F-1> = Ta+5_H2O2_F-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(l):Water 20:80Vol	lgK:1.83(3SF)	0	MSP	6680

Reaction No. 70988							
Ta+3 + 3<e-1> = Ta(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.6(1SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:-0.6(1SF)	4	CRV	22519

Reaction No. 70989							
Ta2O5(s) + 10<H+1> + 10<e-1> = 2<Ta(s)> + 5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.81(2SF)	0	MDG	140
2	25	0	Inf. Dilution	E:-0.750(3SF)	2	CRV	6330
3	25	0	Inf. Dilution	E:-0.750(3SF)	2	CRV	22519
4	25	0.1	Unknown	E:-0.81(2SF)	0	CRV	22551
Note (4): Phase of Ta2O5 not specified in source							

Reaction No. 73376							
Ta2O5(s) + 5<H2O> = 2<Ta+5_OH-1(5)>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-20.(2SF)	1	EES	
Note (1): Estimate based on Peiffer (Nb) and Timofeev (Nb & Ta for high T)							
2	100	3	NaClO4	lgK:-17.4(3SF)	4	MSL	30118
3	150	3	NaClO4	lgK:-17.1(3SF)	4	MSL	30118
4	200	3	NaClO4	lgK:-16.4(3SF)	4	MSL	30118
5	250	3	NaClO4	lgK:-16.4(3SF)	3	MSL	30118

Reaction No. 74263

Ta+5_F-1(7) + 5<e-1> = Ta(s) + 7<F-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-84.44(4SF)	1	ELC	
2	25	0	Unknown	E:-0.45(2SF)	0	CRV	22551

Note (2): Concentration / ionic strength not specified in source

Reaction No. 77489

O2(g) + Ta(s) - e-1 = TaO2+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-201.4(4SF)	0	CRV	28287

Note (1): Species in aqueous solution will be hydrated

Reaction No. 77490

0.5<O2(g)> + Ta(s) - 2<e-1> = TaO+2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-121.65(5SF)	3	CRV	28287

Reaction No. 77494

Ta+2 + 2<e-1> = Ta(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.225(3SF)	3	CRV	28287

Reaction No. 77495

Ta+3 + e-1 = Ta+2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.35(3SF)	3	CRV	28287

Reaction No. 77496

TaO2(s) + e-1 + 4<H+1> = Ta+3 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.8(1SF)	0	CRV	28287

Note (1): Probably, it's Ta2O5(fresh,s) and not Ta2O5(s)

Reaction No. 79586

Ta+5_OH-1(5)_(s) = Ta+5_OH-1(5)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:-5.2(2SF)	3	EOD	403

Note (1): Calc. from Babko solubility [63BLN_30106]							
2	25	1	KNO3	lgK:-5.355(4SF)	3	EOD	7093
Note (2): Calc. from Babko solbility [63BLN_30106]; Ta2O5(s) + 5<H2O> = 2<Ta+5_OH-1(5) > with lgK=-10.71, here divided by 2							

Reaction No. 80122

Ta6O19-8 + H+1 = H+1_Ta6O19-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.02(0.03SD)	2	MGL	7093
Note (1): Application of Bromley theory							
2	25	3	KCl	lgK:13.89(0.02SD)	0	MGL	7093
Note (2): Above MGL limits; pH from 12.4 to 13							
3	25	3	KCl	lgK:11.5(3SF)	2	MMR	30236
Note (3): pH from 10.4 to 14.5							

Reaction No. 80123

Ta6O19-8 + 2<H+1> = H+1(2)_Ta6O19-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.35(0.03SD)	3	MGL	7093
Note (1): Application of Bromley theory							
2	25	3	KCl	lgK:25.91(0.01SD)	1	MGL	7093
Note (2): pH from 12.4 to 13; first pK above MGL limits							

Reaction No. 80124

Ta+5_OH-1(5) = Ta+5_OH-1(4) + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:-13.0(3SF)	3	MSL	30106
2	25	1	KNO3	lgK:-13.0(3SF)	1	EES	

Reaction No. 80125

Ta+5_OH-1(5) + H2O = Ta+5_OH-1(6) + H+1

Note: Basic pK for Ta(OH)5 might not really exist if polyoxotantalates formed

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:-9.6(2SF)	0	MSL	403
2	19	1	KNO3	lgK:-9.6(2SF)	0	MSL	30106
3	25	1	KNO3	lgK:-9.6(2SF)	1	EES	

Reaction No. 80126

Ta+5_OH-1(5) + H+1 = Ta+5_OH-1(4) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:1.(1SF)	3	MSL	403
2	25	1	KNO3	lgK:1.(1SF)	1	EES	

Reaction No. 80127

Ta6O19-8 + H2O = H+1_Ta6O19-8 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:12.68(0.1SD)	3	MGL	30159
Note (1): pH from 8.5 to 12							

Reaction No. 80128

H+1(2)_Ta6O19-8 + H2O = H+1(3)_Ta6O19-8 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:9.28(0.05SD)	3	MGL	30159
Note (1): pH from 8.5 to 12							

Reaction No. 80129

Ta2O5(s) + 6<H+1_F-1> + H2O = 2<Ta+5_F-1(3)_OH-1(3)> + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.(2SF)	1	EDH	
Note (1): Guessing desparately							
2	100	3	NaClO4	lgK:-8.24(3SF)	5	MSL	30118
3	150	3	NaClO4	lgK:-7.45(3SF)	5	MSL	30118
4	200	3	NaClO4	lgK:-8.60(3SF)	5	MSL	30118
5	250	3	NaClO4	lgK:-8.55(3SF)	5	MSL	30118

Reaction No. 80130

Ta+5_F-1(3) + 3<F-1> = Ta+5_F-1(6)

Note: Reacting species unknown, probably hydroxylated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.8(3SF)	1	ELC	
2	25	3	HClO4	lgK:15.7(3SF)	2	MDS	30162

Reaction No. 80131

Ta+5_F-1(3) + 4<F-1> = Ta+5_F-1(7)

Note: Reacting species unknown, probably hydroxylated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.7(3SF)	1	ELC	
2	25	3	HClO4	lgK:20.15(4SF)	2	MDS	30162

Reaction No. 80133

H+1_Ta6O19-8 + H2O = H+1(2)_Ta6O19-8 + OH-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:10.81(0.1SD)	3	MGL	30159
Note (1): pH from 8.5 to 12							

Reaction No. 80134

H+1_Ta6O19-8 + H+1 = H+1(2)_Ta6O19-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.(2SF)	1	EES	
Note (1): Guess based on Ta6O19-8 + 2<H+1> = H+1(2)_Ta6O19-8 (ex Bromley)							
2	25	3	KCl	lgK:9.3(2SF)	4	MMR	30236
Note (2): pH from 10.4 to 14.5							

Reaction No. 80135

3<Cl2(g)> + Cs(s) + Ta(s) = Ta+5_Cs+1_Cl-1(6)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-337.(3SF)	4	CRV	22971

Reaction No. 80136

3<Cl2(g)> + Rb(s) + Ta(s) = Ta+5_Rb+1_Cl-1(6)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-326.(3SF)	4	CRV	22971

Reaction No. 80137

3<Cl2(g)> + K(s) + Ta(s) = Ta+5_K+1_Cl-1(6)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-320.(3SF)	4	CRV	22971

Reaction No. 80138

3<Cl2(g)> + Na(s) + Ta(s) = Ta+5_Na+1_Cl-1(6)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-308.(3SF)	4	CRV	22971

Reaction No. 80139

2<Cl2(g)> + Ta(s) - e-1 = Ta+5_Cl-1(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-149.(3SF)	4	CRV	22971
2	25	0	Inf. Dilution	dH:-170.(3SF)	4	CRV	22971

Reaction No. 80140

1.5<Cl2(g)> + Ta(s) - 2<e-1> = Ta+5_Cl-1(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-123.(3SF)	4	CRV	22971
2	25	0	Inf. Dilution	dH:-139.(3SF)	4	CRV	22971

Reaction No. 80159

6<Ta+5_OH-1(5)> - 6<H+1> = H+1(2)_Ta6O19-8 + 11<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KNO3	lgK:-32.12(4SF)	3	EOD	7093

Note (1): Recalculated own data + solubility Babko (30106)

Reaction No. 80160

6<Ta+5_OH-1(5)> - 7<H+1> = H+1_Ta6O19-8 + 11<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KNO3	lgK:-44.02(4SF)	3	EOD	7093

Note (1): Recalculated own data + solubility Babko (30106)

Reaction No. 80161

6<Ta+5_OH-1(5)> - 8<H+1> = Ta6O19-8 + 11<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KNO3	lgK:-57.48(4SF)	3	EOD	7093

Note (1): Recalculated own data + solubility Babko (30106)

Reaction No. 80169

Ta+5_F-1(3) + 2<F-1> = Ta+5_F-1(5)

Note: Reacting species unknown, probably hydroxylated

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	1	NaClO4	lgK:10.80(4SF)	2	MDS	5398
2	25	3	HClO4	lgK:10.77(4SF)	2	MDS	30162

Reaction No. 80181

Ta+5_OH-1(5) + H+1(4)_EDTA-4 = Ta+5_OH-1(2)_EDTA-4 + 3<H2O> + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1.08	K2SO4	lgK:33.6(3SF)	3	MPL	30359
2	25	0	Inf. Dilution	lgK:34.(2SF)	1	EES	

Reaction No. 80196

Ta+5_OH-1(2)_Oxalic-2 + Oxalic-2 = Ta+5_OH-1(2)_Oxalic-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0	Self Medium	lgK:7.4(2SF)	3	MSL	30107
2	25	0	Inf. Dilution	lgK:7.5(2SF)	1	EES	

Reaction No. 80206

Ta(s) = Ta(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:176.67(5SF)	3	CRV	31291
2	25	0	Inf. Dilution	dH:186.90(5SF)	3	MTD	31291

Reaction No. 80209

Ta+5_OH-1(2)_Oxalic-2(3) = Ta+5_OH-1(2)_Oxalic-2(2) + Oxalic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Self Medium	lgK:5.91(0.05SD)	2	MSL	30164

Note (1): pH 0.3-1.8

Reaction No. 80211

Ta+5 + OH-1 = Ta+5_OH-1

Note: Ta+5 is considered too highly charged to exist

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:29.6(3SF)	0	EQC	30106
Note (1): Highly dubious method							
2	25	1	KNO3	lgK:29.6(3SF)	0	EES	

Reaction No. 80212

Ta+5_OH-1 + OH-1 = Ta+5_OH-1(2)

Note: Ta+5_OH-1 is considered too highly charged to exist

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:26.6(3SF)	0	EQC	30106
Note (1): Highly dubious method							
2	25	1	KNO3	lgK:26.6(3SF)	0	EES	

Reaction No. 80213

Ta+5_OH-1(2) + OH-1 = Ta+5_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:22.4(3SF)	0	EQC	30106
Note (1): Highly dubious method							
2	25	1	KNO3	lgK:22.4(3SF)	1	EES	

Reaction No. 80214

Ta+5_OH-1(3) + OH-1 = Ta+5_OH-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	KNO3	lgK:17.7(3SF)	0	EQC	30106
Note (1): Highly dubious method							
2	25	1	KNO3	lgK:17.7(3SF)	1	EES	

Reaction No. 80525

Ta2O5(s) + 10<H+1_F-1> = 2<Ta+5_F-1(5)> + 5<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.13(2SF)	1	EES	
2	100	3	NaClO4	lgK:0.13(2SF)	5	MSL	30118
3	150	3	NaClO4	lgK:0.35(2SF)	5	MSL	30118

Reaction No. 80596

1.5<O2(g)> + 2<Ta(s)> = Ta2O3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-459.71(5SF)	3	CRV	31291

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EQC	Quintuply charged central ion method
ESC	Standard conditions anchor
MAX	Anion exchange
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MGL	Glass electrode
MIS	Ion selective electrode
MMR	Nuclear magnetic resonance
MPL	Polarography
MQH	Quinhydrone electrode
MSL	Solubility

MSP	**Spectroscopy**
MTD	**Temperature difference**

References

Reference No.: 140(64SiM)
Stability Constants of Metal-ion Complexes SP17, 1964 Sillen LG; Martell AE; The Chemical Soc.,London

Reference No.: 403(76BaM)
The Hydrolysis of Cations, 1976 Baes CF Jr; Mesmer RE; Wiley, New York

Reference No.: 437(80BoH)
IUPAC Chemical Data Series No. 27, 1980 Bond AM; Hefter GT; Critical Survey of Stability Constants and Related Thermodynamic Data of Fluoride Complexes in Aqueous Solution; Pergamon, Oxford, U.K.

Reference No.: 906(79Per)
IUPAC Chemical Data Series No. 22, 1979 Perrin DD; Stability Constants of Metal-Ion Complexes, First Edition Part B, Organic Ligands; Pergamon, Oxford

Reference No.: 931(71SiM)
Stability Constants of Metal-ion Complexes SP25, 1971 Sillen LG; Martell AE; Supplement No 1 to Special Publication No 25; Chemical Society, London

Reference No.: 5398(82Hog)
IUPAC Chemical Data Series No. 21, 1982 Hogfeldt E; Stability Constants of Metal-ion Complexes, 2nd Suppl., Part A, Inorganic Ligands; Pergamon, Oxford

Reference No.: 5766(72Bau)
J. Inorg. Nucl. Chem., 1972, 34, 687 - 695; Baumann EW; Investigation of the Tantalum Fluoride System Using the Fluoride-Selective Electrode

Reference No.: 6330(95LiF)
CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 6422(95BaP)
Thermochemical Data of Pure Substances, 3rd Edn., 1995 Barin I; Platzki G; Weinheim, Germany

Reference No.: 6680(73LCS)
Russ. J. Inorg. Chem., 1973, 18, 533 - 535; Latysh GN; Chernyak AS; Serebrennikova TE; Investigation of Peroxo(fluoro)-complexes of Niobium and Tantalum in Sulphuric Acid Solutions

Reference No.: 7093(95AEF)
J. Solution Chem., 1995, 24, 611 - 622; Arana G; Etxebarria N; Fernandez LA; Madariaga JM; Hydrolysis of Nb(V) and Ta(V) in Aqueous KCl at 25 C. Part II. Construction of a Thermodynamic Model for Ta(V)

Reference No.: 21702(62VaF)
J. Phys. Chem., 1962, 66, 187 - 189; Varga LP; Freund H; The Formation Constants of the Tantalum Fluoride System. II. Tantalum Electrode Potential Studies

Reference No.: 22519(05Lid)
CRC Handbook of Chemistry and Physics, 86th Edn., 2005 Lide DR; A Ready-Reference Book of Chemical and Physical Data; Taylor & Francis, Boca Raton, USA

Reference No.: 22551(06Inz)
Encyclopedia of Electrochemistry, Vol. 7a, 2006, 17 - 75; Inzelt G; Standard, Formal and Other Characteristic Potentials of Selected Electrode Reactions

Reference No.: 22971(71HWH)
Chem. Rev., 1971, 71, 127 - 137; Hill JO; Worsley IG; Hepler LG; Thermochemistry and Oxidation Potentials of Vanadium, Niobium and Tantalum

Reference No.: 28287(92IPJ)
J. Phys. Chem., 1992, 96, 11096 - 11101; Ionova GV; Pershina V; Johnson E; Fricke B; Schadel M; Redox Reactions for Group 5 Elements, Including Element 105, in Aqueous Solutions

Reference No.: 30106(63BLN)
Russ. J. Inorg. Chem., 1963, 8, 957 - 961; Babko AK; Lukachina VV; Nabivanets BL; Solubility and Acid-Base Properties of Tantalum and Niobium Hydroxides

Reference No.: 30107(65BLN)
Russ. J. Inorg. Chem., 1965, 10, 467 - 474; Babko AK; Lukachina VV; Nabivanets BL; Oxalato-Complexes of Tantalum

Reference No.: 30109(66BFH)
J. Inorg. Nucl. Chem., 1966, 28, 421 - 431; Bukhsh MN; Flegenheimer J; Hall FM; Maddock AG; de Miranda CF; The Chemistry of Protactinium - VII. The Fluoro-Complexes

Reference No.: 30118(17TMW)
Geochim. Cosmochim. Acta, 2017, 197, 294 - 304; Timofeev A; Migdisov AA; Williams-Jones AE; An Experimental Study of the Solubility and Speciation of Tantalum in Fluoride-bearing Aqueous Solutions at Elevated Temperature

Reference No.: 30139(54Hum)
J. Am. Chem. Soc., 1954, 76, 978 - 980; Humphrey GL; Heats of Formation of Tantalum, Niobium and Zirconium Oxides, and Tantalum Carbide

Reference No.: 30159(69Spi)
Compt. Rend. Ser. C. Chim., 1969, 269, 1108 - 1111; Spinner B; New Tantalum(V) Isopolyanions

Reference No.: 30162(65VWN)
Anal. Chem., 1965, 37, 1003 - 1009; Varga LP; Wakley WD; Nicolson LS; Madden ML; Patterson JH; Solvent Extraction Studies of Tantalum Fluoride Complexes with N-Benzoylphenylhydroxylamine, Tri-N-octylphosphine Oxide and Methyl Isobutyl Ketone using Computer Techniques

Reference No.: 30164(70ZhP)
Sov. Radiochem. (Engl. Transl.), 1970, 12, 89 - 94; Zhurenkov EM; Pobezhimovskaya DN; Determination of the Composition and Instability Constants of Oxalate Complexes of Niobium and Tantalum by the Solubility Method

Reference No.: 30236(07BAR)
Inorg. Chem., 2007, 46, 7032 - 7039; Balogh E; Anderson TM; Rustad JR; Nyman M; Casey

WH; Rates of Oxygen-Isotope Exchange between Sites in the [HxTa6O19](8-x)-(aq) Lindqvist Ion and Aqueous Solutions: Comparisons to [HxNb6O19](8-x)-(aq)

Reference No.: 30359(69VoS)
Russ. J. Inorg. Chem., 1969, 14, 652 - 654; Volkova GA; Sochevanov VG; Reaction of Tantalum Hydroxide with Ethylenediaminetetra-acetic Acid

Reference No.: 30601(66StA)
Vestn. Leningr. Univ., Fiz., Khim., 1966, 21, 133 - 138; Stolyarov KP; Amantova IA; Photometric Study of Ascorbic Acid Complexes. VI. The Systems Zr, Hf, V(IV), Nb, Ta, Mo, W - Ascorbic acid

Reference No.: 31291(84PSG)
U.S. Bur. Mines Bull. 677, 1984 Pankratz LB; Stuve JM; Gokcen NA; Thermodynamic Data for Mineral Technology; US Det. Interior, Bur. Mines, Washington, DC, USA